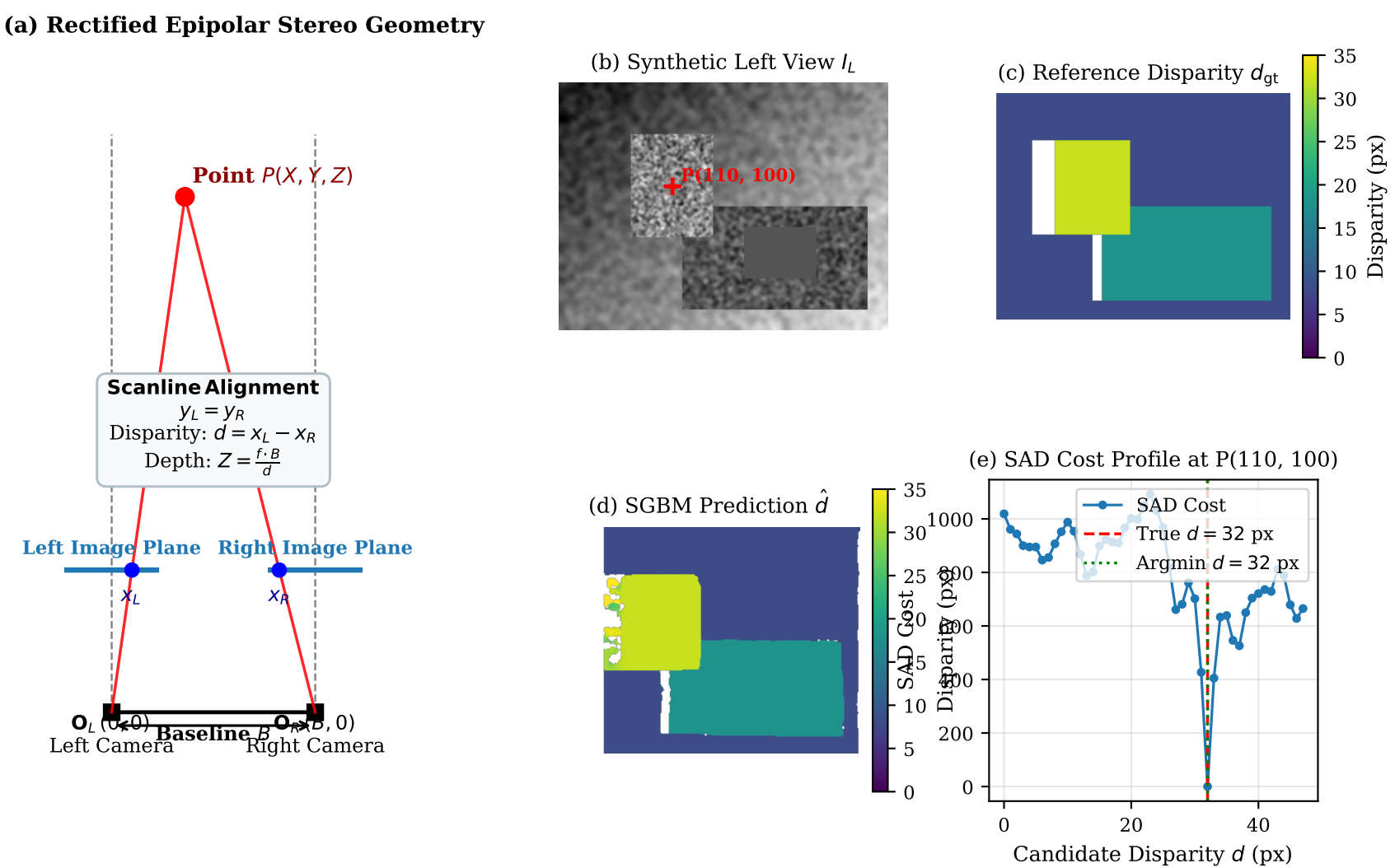

(a) Rectified Epipolar Stereo Geometry
Point $P(X, Y, Z)$
Scanline Alignment
$y_L = y_R$
Disparity: $d = x_L - x_R$
Depth: $Z = \frac{f \cdot B}{d}$
Left Image Plane
Right Image Plane
x_L
x_R
O_L (0, 0)
Baseline B
O_R (B, 0)
Left Camera
Right Camera
(b) Synthetic Left View I_L
P(110, 100)
(c) Reference Disparity d_{gt}
Disparity (px)
(d) SGBM Prediction $\hat{d}$
SAD Cost
(e) SAD Cost Profile at P(110, 100)
SAD Cost
True $d = 32$ px
Argmin $d = 32$ px
Disparity (px)
Candidate Disparity d (px)